Reg. No. | | | | | | | | | | | [14CE43]

FOURTH SEMESTER B.Tech. EXAMINATION, JUNE – 2016

GEOLOGY

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions choosing at least any two full questions from each part.
2. Missing data may be suitably assumed.

PART - A

1. a. Define Geology. 01
   b. Discuss the importance of Geology in the field of Civil Engineering. 05
   c. Define different important branches of Geology and their application to Civil Engineering in planning, design, construction and execution? 08

2. a. Define Mineral. 02
   b. Describe the physical properties, chemical composition, uses and Engineerng importance of the following Minerals. 12
      i. Orthoclase
      ii. Galena
      iii. Hematite

3. a. What is Rock? 02
   b. What is Monomineralic and Multimineralic rock? Explain with example. 04
   c. Explain the classification of Igneous rocks with suitable example and mention the engineering importance of Igneous rocks. 08

4. Write the properties and uses of the following. 14
   i. Granite
   ii. Basalt
   iii. Sand stone
   iv. Marble

PART – B

5. a. What is Fault? 02
   b. Describe the parts of Faults. 04
   c. Describe the types of Faults. 08

6.	**a.**	Define out crop.	**03**
	b.	Define Dip and strike	**05**
	c.	Comment on tunnelling through the folded strata.	**06**
7.	**a.**	What is porosity of Rocks?	**05**
	b.	What is permeability of Rocks?	**04**
	c.	Write a brief note on preventive measures of sitting of the Damn/Reservoirs.	**05**
8.	**a.**	What is Ground Water Table? What is the effect of release of strain energy in rocks?	**04**
	b.	Explain occurrence of Ground Water (vertical distribution of Ground water)	**10**